Transfer: gravity and motion

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Apply gravity, projectile and orbital reasoning to a new set of cases.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 13.*
Velocity tangent; acceleration inward at both positions. Constant speed does not mean constant velocity.

02. Worked example: explain the model
Both axes evolve over the same 2 s; horizontal distance 8 m, vertical motion governed by gravity.

03. Read and interpret the model
Tangent v, inward gravity; shared freefall removes support force.

04. Apply the physics
12 m; new 6 m.

05. Explain your reasoning
Need time phase/geometry/Earth position/uncertainty; valid specific request.

06. Review: connect an earlier idea *Review: Unit 2, Day 4.*
4 N; divide by 36 because force follows inverse square separation.

07. Exit check
g 2 N/kg; force 6 N inward.

08. STRETCH: test the model
Triple horizontal speed; keep height, initial vertical velocity and gravitational field fixed, with negligible air resistance.

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01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 13.*
Speed becomes 1/3; period becomes 27 times as large. $v \propto r^{(-1/2)}$, $T \propto r^{(3/2)}$.

02. Worked example: explain the model
Both axes evolve over the same 2 s; horizontal distance 8 m, vertical motion governed by gravity.

03. Read and interpret the model
Tangent v, inward gravity; shared freefall removes support force.

04. Apply the physics
t 2 s; range 12 m; v_y=-20 m/s; speed $\sqrt{436}$=20.9 m/s.

05. Explain your reasoning
Need time phase/geometry/Earth position/uncertainty; valid specific request.

06. Review: connect an earlier idea *Review: Unit 2, Day 4.*
Fixed masses: 4 r. Both masses enlarged: 24 r gives the required force fraction.

07. Exit check
v 4 m/s; a=0.64 m/s²; T 39.3 s; direction changes while speed stays constant.

08. STRETCH: test the model
t=3 s, range 36 m, velocity (12,-30) m/s. At half g, time and range multiply by $\sqrt{2}$; new range≈50.9 m.